# From Ideation to Production in 7 days

## The Scoring Factory at Raiffeisen

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RAIFFEISEN

# Who we are

RAIFFEISEN

- Third largest bank in Switzerland
- Cooperative organisation with 226 independent regional units
- Retail bank

## Data Science

- Since 2016
- Extract and deliver information to client advisors, marketing, sales and other business units
- Infer patterns from customer data (3.5 mio) and model customer behavior and needs

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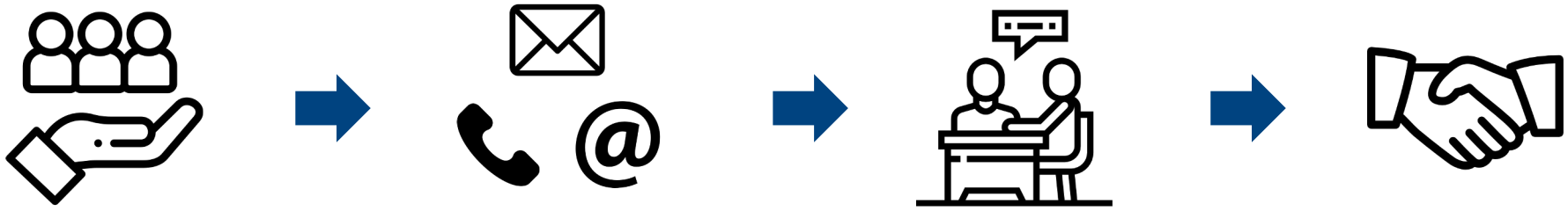
- Consultancy for data-driven value creation
- Curation of the most talented data team in Switzerland
- 100+ Consultants in Zurich
- 20+ in Athens
- Founded 2005, 1’000+ data projects
- International and Swiss clients
- Selectively invested in startups

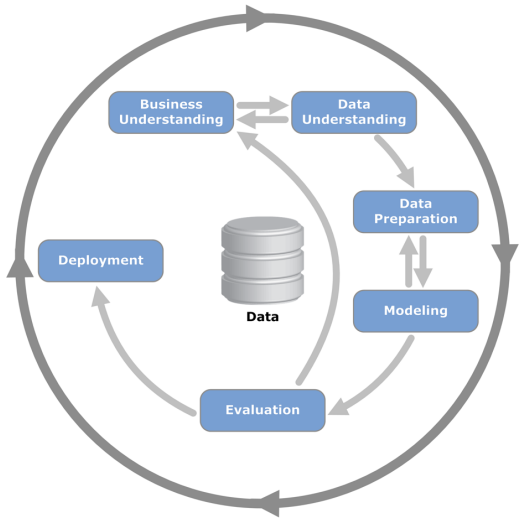
# How does sales analytics generate value?



Customer selection (Analytics) → Contact → Appointment with client advisor → Deal

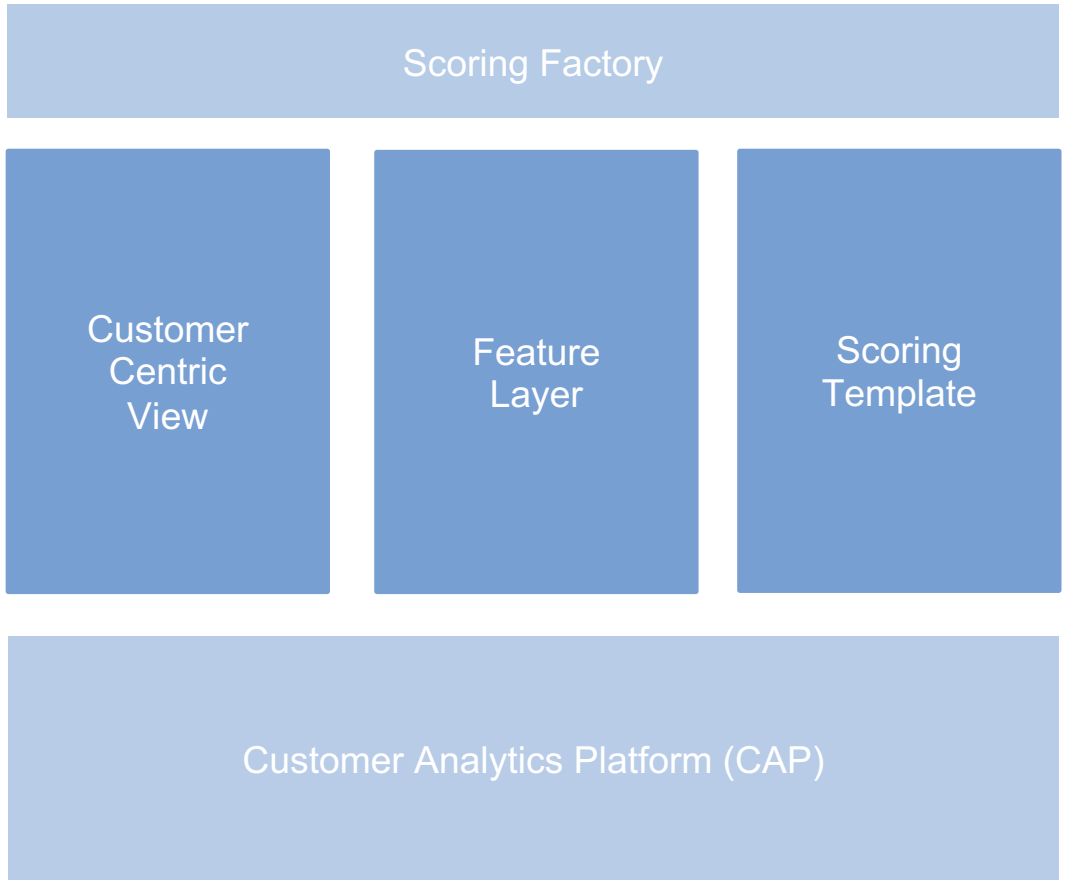
- Impact of good selection: efficient allocation of time and money
- Limited time availability & budget (Data Scientists and client advisors)

# Data Science process



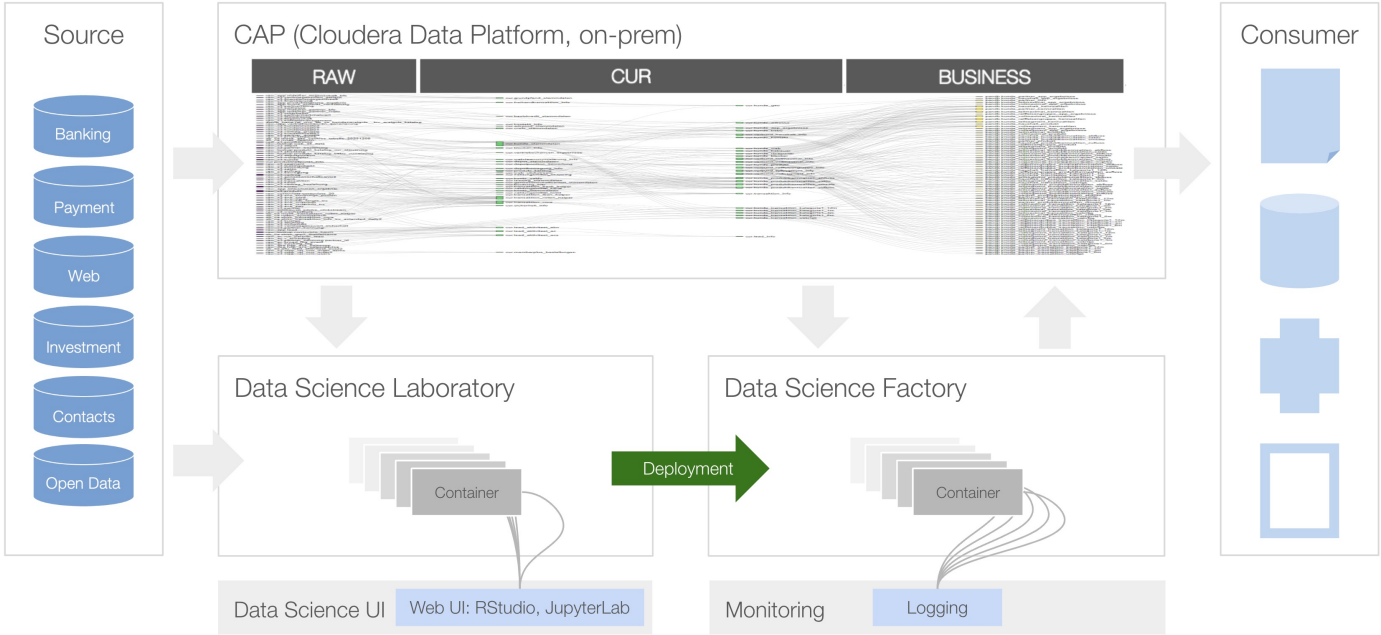

- Time to prototype several months
  → High upfront costs for every idea
- Manual recycling of old code

# Solution: 3 Pillars

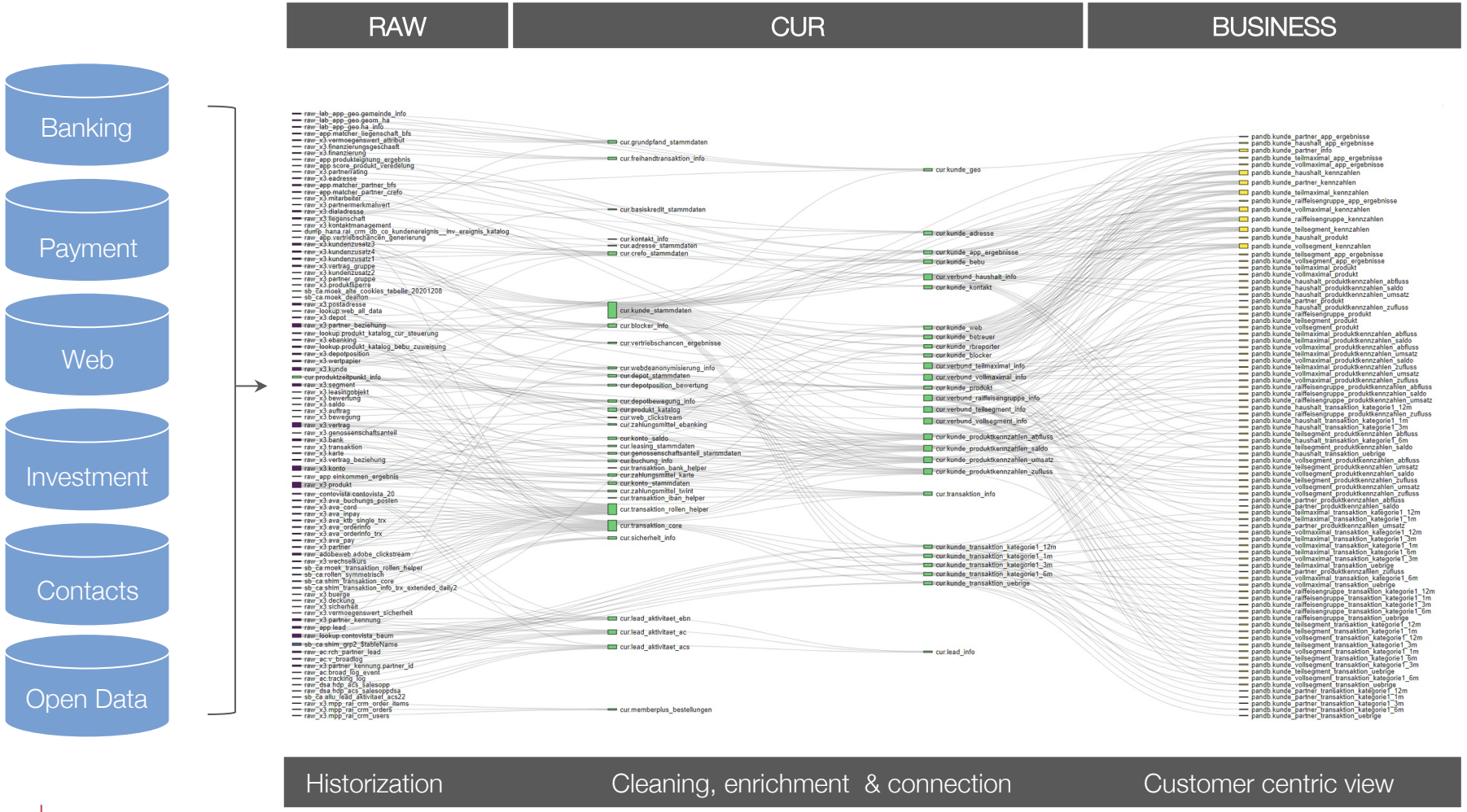


# Foundation: Customer Analytics Platform (CAP)

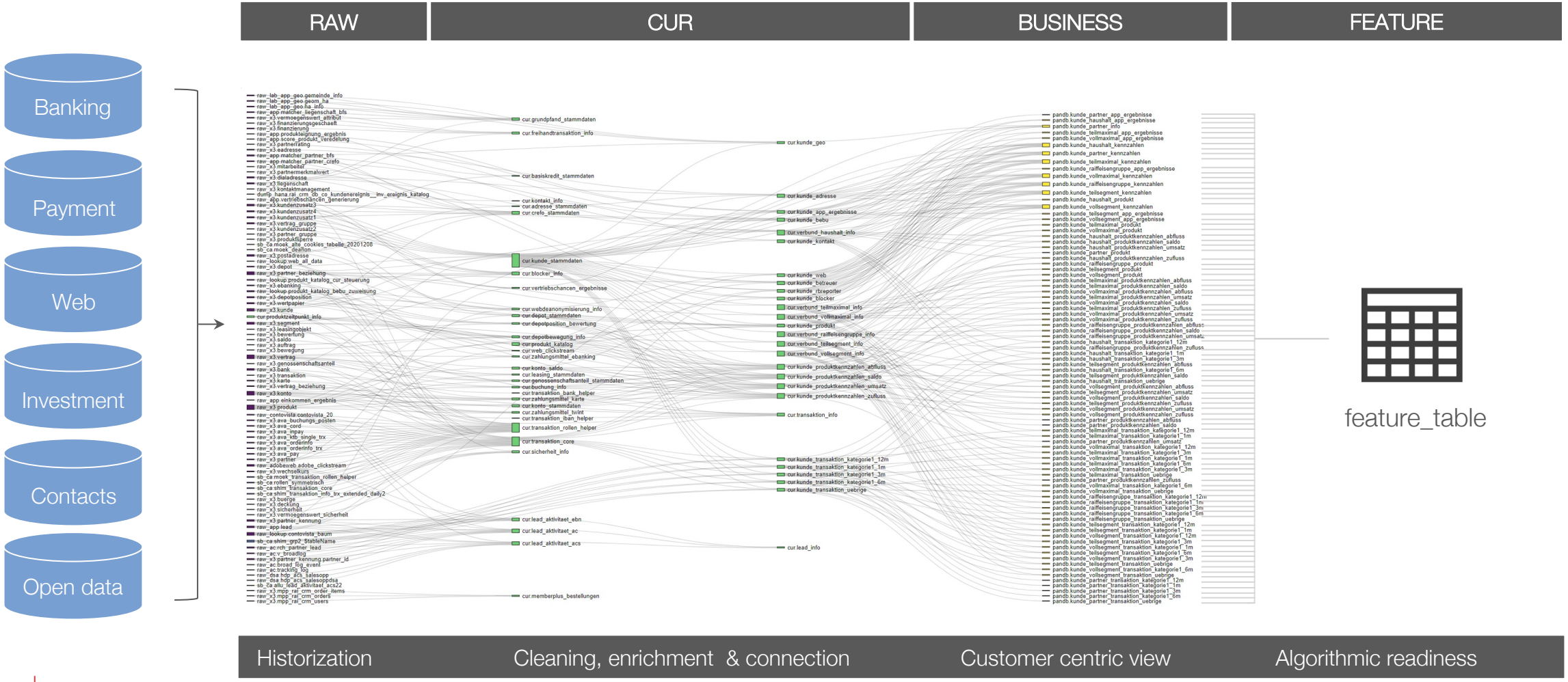
- 100+ Users
  - Business Analysts, Data Scientists
  - Customer Analytics, Marketing, Risk
- Data Science
  - R, Python, SQL
- Technology
  - Cloudera Data Platform
  - Lab-UI and Factory running in project-specific Docker Containers
  - Azure DevOps Server



# Pillar 1: Customer centric view



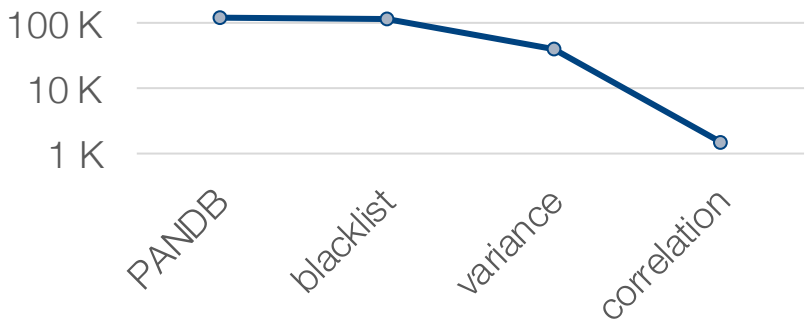
# Pillar 2: The feature layer



# Pillar 2: Data prep in feature layer

3 step **feature reduction**: Reduction of amount and redundancy of the customer data

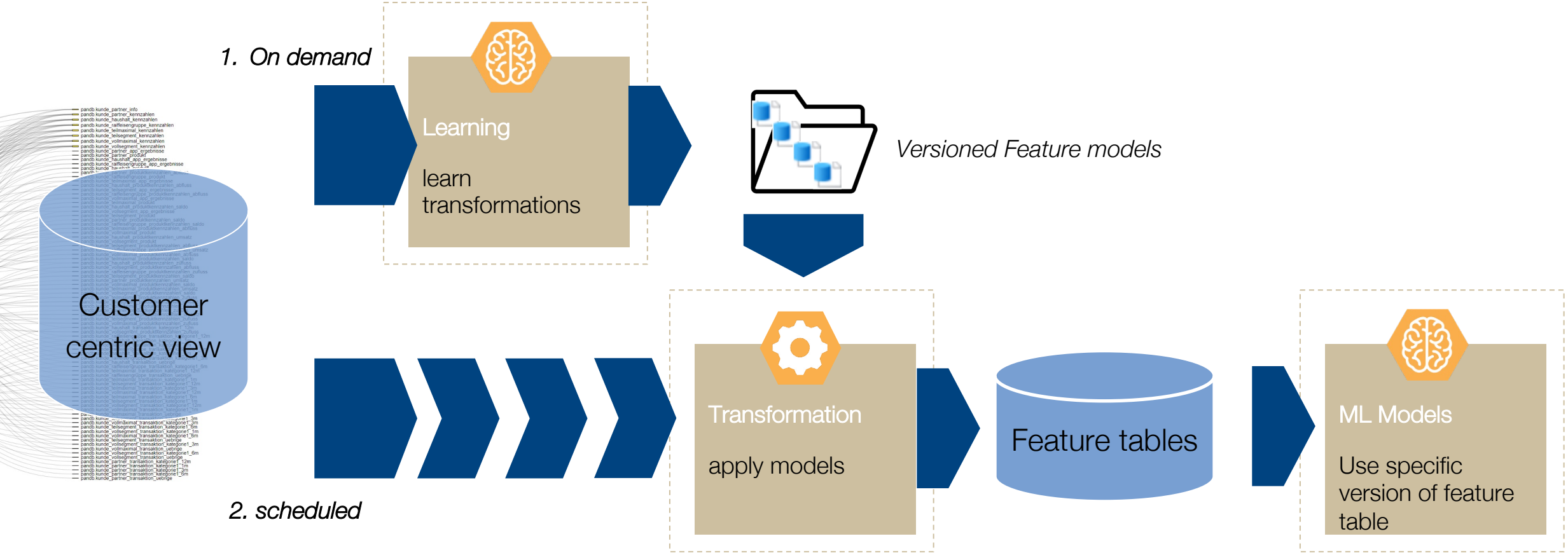
- Technical attributes: Blacklist
- Variance: Filter attributes with low variance
- correlation: Remove redundant, correlated attributes




Pre-defined, consistent and flexible model to **treat missing values**

**one-hot-encoding** of nominal attributes

# Pillar 2: The architecture of the feature layer



# Pillar 3: Scoring template

Code shell

- Production ready structure
- Modular
- Automated
- Consistent

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Use case specific

data loading

- Labels
- Eligible customers
- Features

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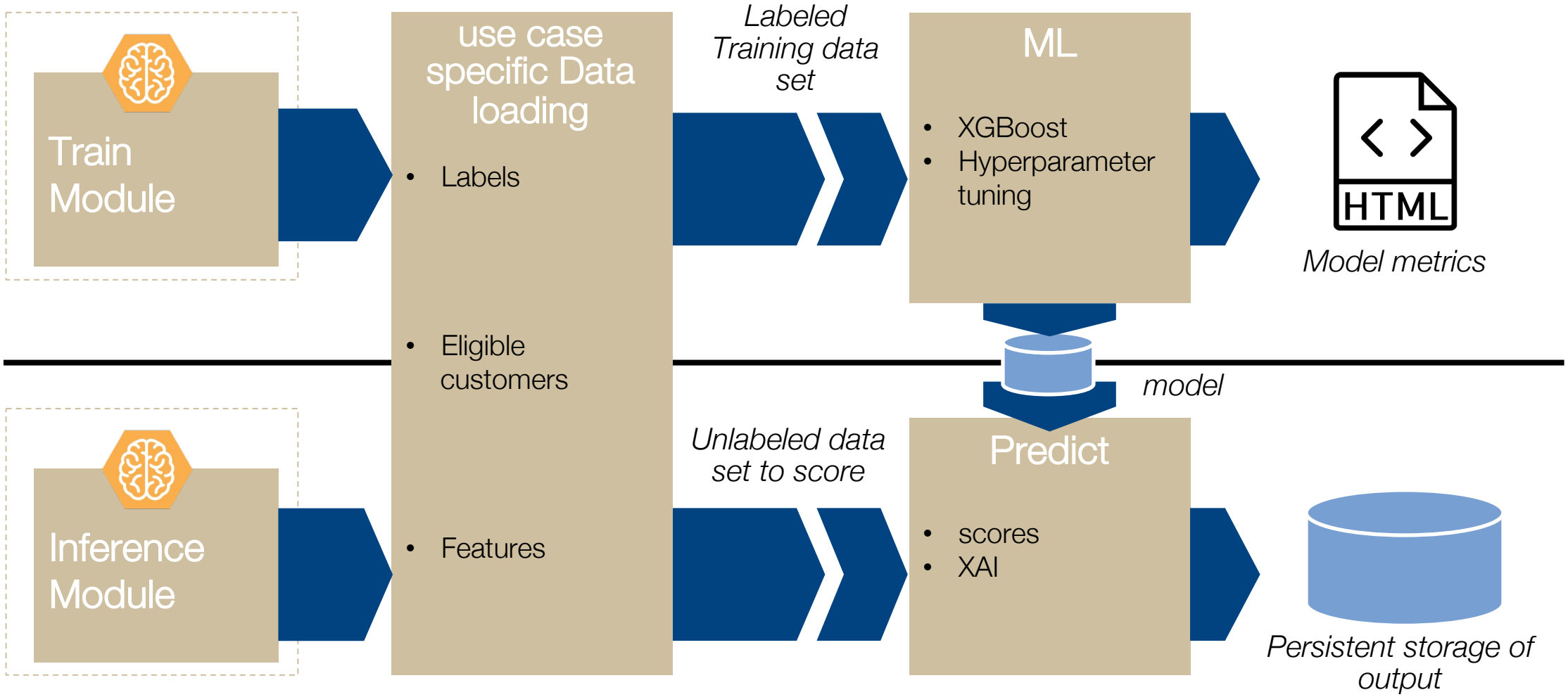
Production ready

ML Model

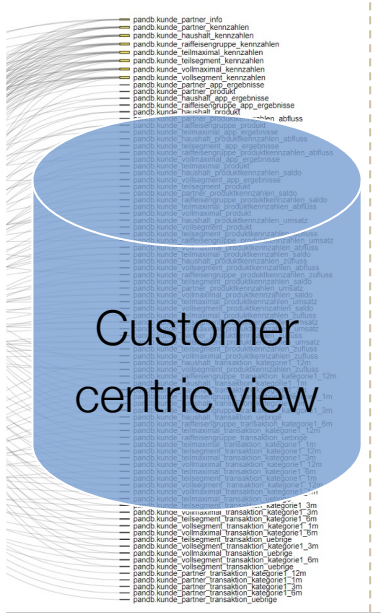
Automation and abstraction of all technicalities

business relevant questions

# Pillar 3: Data flow in the scoring template



# Use Case: Product recommendation




## Feature layer

- Algorithmic readiness

## Product score

- Predict sales probabilities for every product and customer.
- SHAPly explanations

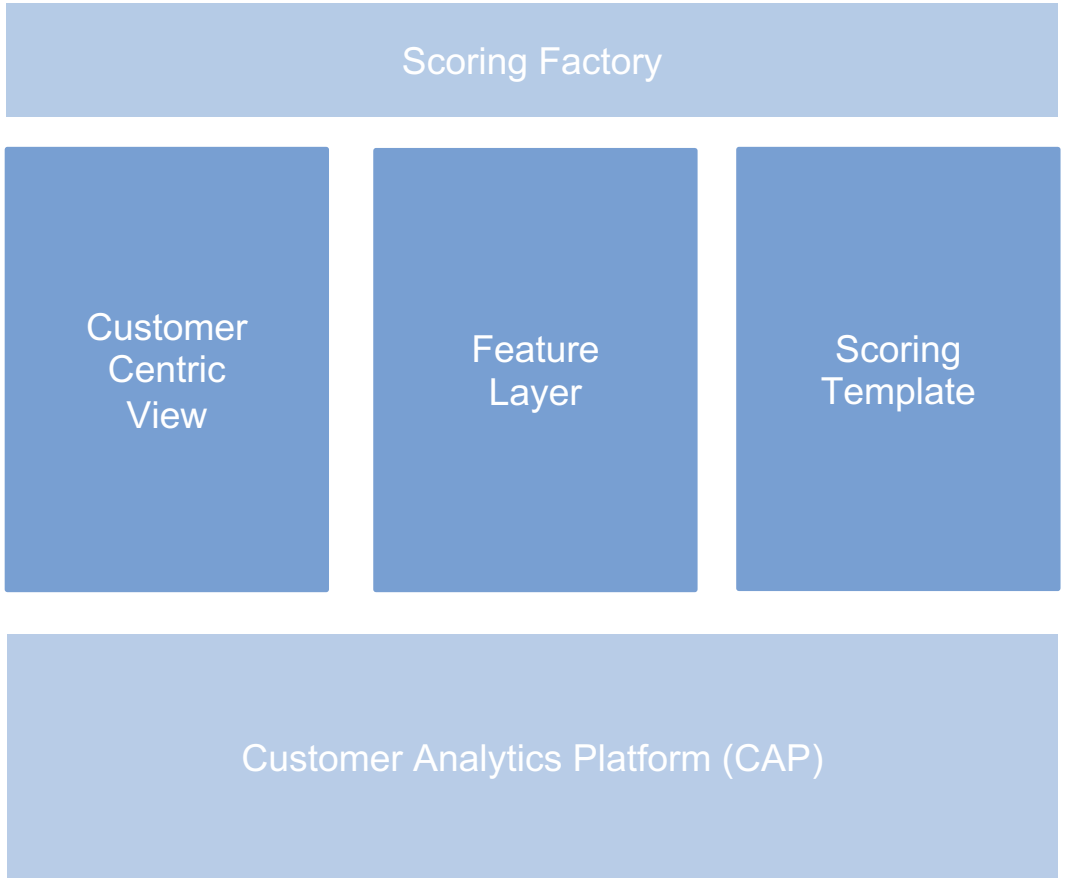
## Business rules

- Reweight sales probabilities with business value
- Up to 3 product recommendations per customer

## Lead management

- Consistency and compatibility checks
- Deliver to core banking system

# Impact




- Reduced implementation time and upfront costs
  → From Idea to production in ~1 week
- Business drives use cases (instead of data scientists)
- Increase in conversion rate of factor 3 for leads from product scores

# Thank you for your attention!

Contact us:

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